Method	Unified Affine Form	Activation Impact ($\Delta\mathbf{h}$)	Param. Size
Local Weight	$(\mathbf{W} + m\,\Delta\mathbf{W})\,\mathbf{h}_i + (\mathbf{b} + m\,\Delta\mathbf{b})$	$m\,(\Delta\mathbf{W}\,\mathbf{h}_i + \Delta\mathbf{b})$	$d_{\text{in}} \times d_{\text{out}} + d_{\text{out}}$
LoRA	$(\mathbf{W} + m\,\mathbf{BA})\,\mathbf{h}_i + \mathbf{b}$	$m\,(\mathbf{BA}\,\mathbf{h}_i)$	$d_{\text{in}} \times r + r \times d_{\text{out}}$
Steering Vector	$\mathbf{W}\,\mathbf{h}_i + (\mathbf{b} + m\,\Delta\mathbf{b})$	$m\,\Delta\mathbf{b}$	d_{out}

Table 1: All methods in our unified framework, expressed under the affine weight-update formulation and their corresponding activation changes $\Delta\mathbf{h}$. d_{in} and d_{out} denote the input and output dimensions of the layer; r is the LoRA rank with $r \ll \min(d_{\text{in}}, d_{\text{out}})$.

preference is increased at the expense of utility, completions often become incoherent or instruction-violating, reducing usability and obscuring the intended concept signal under output-based evaluation. Therefore, effective model control should shift preference while preserving utility.

Notation. Given a query q, we construct a polarity pair of completions: a concept-positive answer A_p and a concept-negative answer A_n. We denote their conditional probabilities as $P(A_p \mid q)$ and $P(A_n \mid q)$, and define the corresponding cross-entropy losses as $\mathcal{L}_p \triangleq -\log P(A_p \mid q)$ and $\mathcal{L}_n \triangleq -\log P(A_n \mid q)$. We further introduce latent preference probabilities $P(p_p \mid q)$ and $P(p_n \mid q)$, as well as a polarity-invariant task-success probability $P(u \mid q)$.

Preference–Utility Factorization. Following prior work that assumes concept directions are mutually orthogonal, we likewise treat concept preference as independent from task utility for a given query q. Concretely, for a polarity pair (A_p, A_n), we decompose

$$\begin{aligned} P(A_p \mid q) &= P(u \mid q)\, P(p_p \mid q), \\ P(A_n \mid q) &= P(u \mid q)\, P(p_n \mid q), \end{aligned} \tag{6}$$

where $P(u \mid q)$ is shared across the pair and $P(p_p \mid q) + P(p_n \mid q) = 1$.

Preference Log-odds. The shared utility cancels in the likelihood ratio, yielding

$$\mathrm{PrefOdds}(q) \triangleq \log \frac{P(p_p \mid q)}{P(p_n \mid q)} = \mathcal{L}_n - \mathcal{L}_p. \tag{7}$$

Utility Log-odds. The total probability mass assigned to the matched pair recovers utility, $P(u \mid q) = P(A_p \mid q) + P(A_n \mid q)$; substituting $P(A \mid q) = e^{-\mathcal{L}}$ gives

$$\begin{aligned} \mathrm{UtilOdds}(q) &\triangleq \log \frac{P(u \mid q)}{1 - P(u \mid q)} \\ &= \log \frac{e^{-\mathcal{L}_p} + e^{-\mathcal{L}_n}}{1 - e^{-\mathcal{L}_p} - e^{-\mathcal{L}_n}}. \end{aligned} \tag{8}$$

We use $\mathrm{PrefOdds}$ and $\mathrm{UtilOdds}$ throughout to track how interventions shift concept preference versus task utility on a common additive scale, with additional derivations in Appendix C.

3.3 Unified Dynamics Observation

Experimental Setup. We evaluate dynamic interventions on two types of tasks: (i) a personality tendency classification task (*Psychopathy*), and (ii) open-ended generation using *PowerSeeking* and the top 10 concept subsets from *AxBench*. We run experiments on `Gemma-2-9B-IT` at layer 20 and `Qwen-2.5-7B-Instruct` at layer 14, following Bigelow et al. (2025), and consider three intervention types: local weight, low-rank adaptation LoRA, and vector. We train each type using both the SFT objective and the RePS objective. Additionally, for vector, we include a train-free method called DiffMean (Marks and Tegmark, 2023). More details are provided in Appendix A.

Metrics. For each query q with matched answers (A_p, A_n), we compute preference and utility using the log-odds in Eqs. (7) and (8). These metrics allow us to track how preference and utility evolve as we vary the intervention scale m.

Unified Dynamics. Experimental results show that, under the unified perspective framework, different intervention forms exhibit remarkably consistent dynamic patterns. As shown in Figure 2, localized weight updates, low-rank adaptation (LoRA), and vector-based interventions display highly similar overall curve shapes. Additional results are included in Appendix A.

For preference log-odds, all methods typically follow a three-stage pattern when plotted against the steering factor m: for small $|m|$, they enter a *Linear Region*, where log-odds grows approximately linearly with m (Bigelow et al., 2025); this is followed by a *Transitional Region* with a noticeable change in trend, and finally a *Convergence Region* where the curve flattens and stabilizes.

Utility log-odds, in contrast, generally peak near

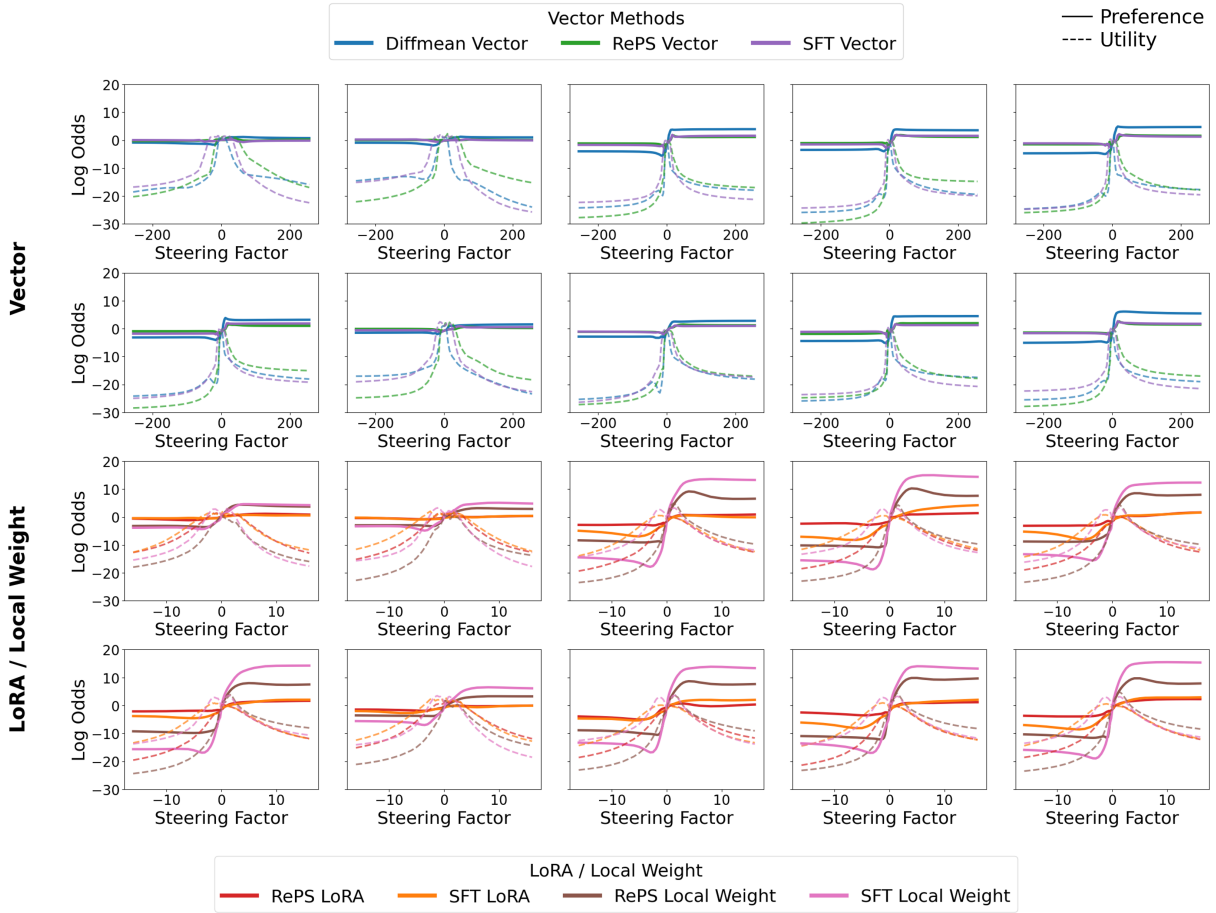

Figure 2: **Unified preference and utility dynamics under steering.** Solid lines represent preference log-odds, and dashed lines represent utility log-odds. The top panel shows steering with vector-form parameter modifications, and the bottom panel shows parametric interventions including LoRA and local weight updates. Results are shown for the `Gemma-2-9B-IT` model on the *AxBench* dataset, evaluated over its top 10 concept subsets. The horizontal axis corresponds to the steering factor.

$m \approx 0$, and remain near their maximum within this narrow range. As $|m|$ increases, utility gradually declines and eventually stabilizes.

These patterns reveal a unified steering response of preference and utility.

4 Capability Dynamics: Mechanism Analysis and Optimization

Motivated by the unified preference–utility dynamics observed across intervention forms (Figure 2), this section provides a mechanistic account and an empirical characterization. We take an activation-manifold perspective and introduce a simple validity-decay factor to capture the tendency for capability to degrade as steering pushes activations away from the activation manifold, without committing to a specific underlying geometry. On this basis, we express preference as the combined effect of (i) steering-induced preference projection changes and (ii) validity decay, while utility is modeled as being dominated by the validity decay term. Finally, under this hypothesis we formalize how the steering factor m shapes both preference and utility log-odds, and show via curve-fitting that the resulting forms match the observed log-odds–m dynamics well across settings.

4.1 Activation Manifold Hypothesis

Prior work suggests that model activations often concentrate on low-dimensional, manifold-like sets in representation space (Bricken et al., 2023; Wollschläger et al., 2025). Adopting this manifold perspective, we analyze additive steering as a translation of hidden states along an approximately fixed direction in activation space. Intuitively, small translations may adjust model behavior in a targeted way, whereas large translations may push representations away from the high-density region learned during training, increasing the risk of a

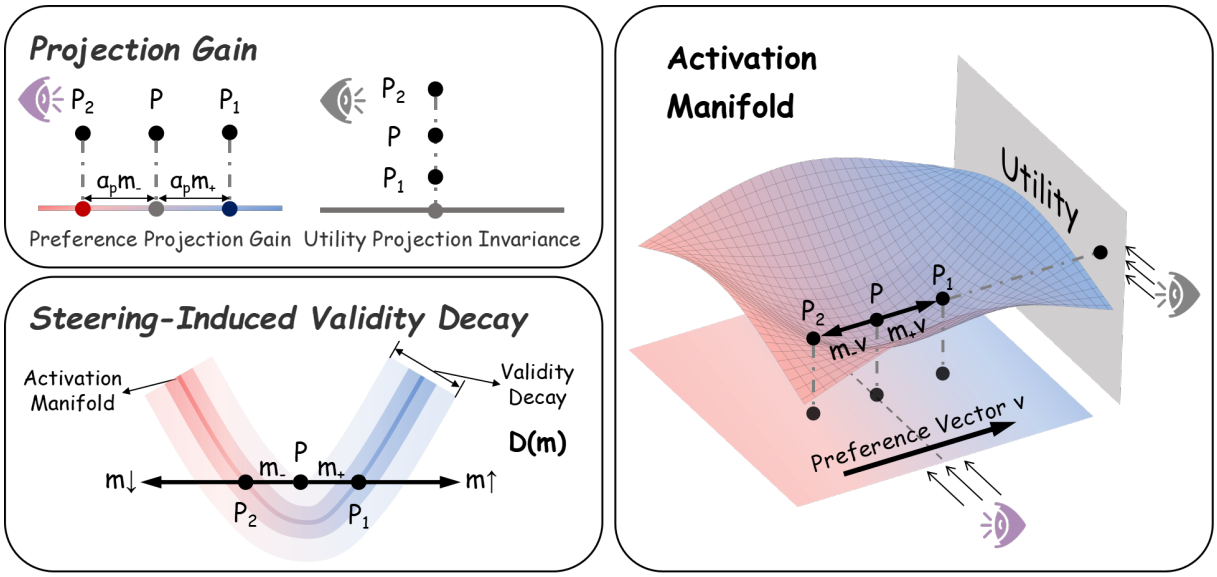

Figure 3: **Mechanism of projection gain and validity decay. Right:** An activation manifold view illustrating Assumption 4.1. An activation P lies on or near the manifold. Steering using preference vector v with scaling factors m_+ and m_- moves P to P_1 and P_2, corresponding to intersections with the manifold. **Top-left:** Projection gain. Projections onto the utility axis exhibit limited variation, whereas projections along the preference direction differ between P_1 and P_2, suggesting that steering primarily influences preference-related components. **Bottom-left:** Steering-induced validity decay. As assumed in Assumption 4.2, increasing steering factor increases off-manifold deviation, leading to a monotonic decrease in validity and degraded downstream decoding.

representation–decoder mismatch and thus degrading general capability.

We formalize this view with two assumptions.

Assumption 4.1 (Training-Induced Activation Manifold). Fix a layer l with hidden dimension d_l. There exists a low-dimensional set (or its neighborhood) $\mathcal{M}_l \subset \mathbb{R}^{d_l}$ such that for inputs x drawn from a set of *stably handled* inputs $\mathcal{X}_{\text{stable}}$, the corresponding activation $h_l(x)$ lies on or near $\mathcal{M}_l$ with high probability:

$$\Pr_{x\sim\mathcal{X}_{\text{stable}}} [d(h_l(x), \mathcal{M}_l) \leq \epsilon] \geq 1 - \delta, \quad (9)$$

where $d(\cdot, \mathcal{M}_l)$ denotes distance to $\mathcal{M}_l$, $\epsilon > 0$ is a neighborhood radius, and $\delta \in (0, 1)$ is a small failure probability.

Assumption 4.1 asserts that pre-training induces a "typical" region of activation space where representations concentrate for stably handled inputs. We next introduce a generic notion of *representation validity*, which is high near $\mathcal{M}_l$ and decreases as hidden states move away from it. This abstraction avoids committing to a specific geometry for $\mathcal{M}_l$ while retaining the key implication: sufficiently off-manifold activations are more likely to be decoded unreliably by the remaining network.

Assumption 4.2 (Steering-Induced Validity Decay). Let $F_{l\to L}$ denote the remainder of the model from layer l to the output logits. There exists a *validity* function $V_l : \mathbb{R}^{d_l} \to [0, 1]$ that is monotonically non-increasing in $d(h, \mathcal{M}_l)$, capturing how well $F_{l\to L}$ can stably decode an activation h. For an additive steering intervention at layer l,

$$\tilde{h}_l(m) = h_l + m\,\Delta h, \quad (10)$$

with steering direction Δh and steering factor $m \in \mathbb{R}$, define the average validity over stably handled inputs:

$$D(m) \triangleq \mathbb{E}_{x\sim\mathcal{X}_{\text{stable}}}\Big[V_l\Big(\tilde{h}_l(m)\Big)\Big]. \quad (11)$$

We assume that $D(m) \in [0, 1]$ decreases with $|m|$ (i.e., larger interventions induce larger off-manifold shifts on average), and that the resulting capability degradation is dominated by this validity decay.

To connect Assumptions 4.1–4.2 to a concrete functional form, we view steering as moving an